Cooperative Strategy Evolution for Multi-Agent Systems via Post-Adversarial Fine-Tuning in Simulation Environments

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Abstract

The deployment of multi-agent systems in competitive domains such as cyber-defense, financial markets, and strategic wargaming necessitates robust and adaptive cooperative strategies. A key challenge lies in enabling agents to rapidly evolve their policies in response to discovered vulnerabilities or novel adversarial tactics encountered during operations. Simulation wargaming environments offer a controlled, cost-effective platform for extensive adversarial training and subsequent strategy refinement. However, effectively translating adversarial experiences into stable, generalized policy improvements remains an open problem. This paper proposes a novel cyclical framework for cooperative strategy evolution in multi-agent systems, centered on post-adversarial fine-tuning within simulation wargaming environments. The framework consists of three core phases: adversarial confrontation in a high-fidelity simulator, automated vulnerability assessment and trajectory sampling, and targeted policy fine-tuning using advanced multi-agent reinforcement learning (MARL) techniques. We formulate the problem as a meta-game where agents must adapt their cooperative policy based on outcomes from a population of adversarial strategies. The fine-tuning mechanism integrates a stabilized experience replay from critical adversarial episodes and employs a dynamics-aware policy gradient to ensure coherent multi-agent adaptation. Extensive experiments in diverse competitive simulation scenarios demonstrate that our framework enables agents to consistently close performance gaps exposed by adversaries, leading to evolved strategies with significantly enhanced robustness and win rates against adaptive opponents, compared to standard self-play or static training paradigms. The work provides a systematic, simulation-driven pathway for continual multi-agent strategy evolution.

Keywords : Multi-agent systems, strategy evolution, adversarial fine-tuning, simulation wargaming, multi-agent reinforcement learning, policy adaptation

1. Introduction

The application of multi-agent systems (MAS) to complex competitive domains has seen remarkable growth, driven by advances in artificial intelligence [1]. In scenarios such as automated cyber operations, algorithmic trading, and large-scale strategic simulations (wargames), the success of a team of agents hinges not only on their initial cooperative policy but also on their capacity for rapid, post-deployment adaptation. Real-world adversaries are inherently adaptive, exploiting weaknesses in static strategies. Therefore, the ability for a multi-agent team to evolve its cooperative strategy after encountering and analyzing adversarial challenges is critical for long-term effectiveness.

High-fidelity simulation wargaming environments provide an indispensable testbed for this challenge [2]. They allow for the generation of vast, diverse adversarial interactions without real-world costs or risks. The core research problem shifts from learning a single, optimal policy to designing a process or mechanism for continual strategy evolution. Specifically, how can a team of agents, after an adversarial engagement within a simulator, identify the root causes of failure or suboptimal performance, and then efficiently fine-tune their joint policy to address these specific vulnerabilities? This process of post-adversarial fine-tuning must be stable, sample-efficient, and preserve previously acquired cooperative skills.

Current MARL paradigms, including centralized training with decentralized execution (CTDE) and population-based training [3], [4], offer a foundation. However, they often lack a structured, automated pipeline for focused policy refinement following adversarial encounters. Self-play [5], while powerful, can be slow to discover specific strategic vulnerabilities and may not efficiently allocate learning resources to the most critical failure modes.

To address this gap, this paper proposes a novel cyclical framework for Cooperative Strategy Evolution via Post-Adversarial Fine-Tuning (CSE-PAFT). The main contributions are:

1. A Formalized Evolution Cycle: We introduce a three-phase cyclic framework (Adversarial Confrontation, Vulnerability Assessment & Sampling, Targeted Fine-Tuning) that formalizes the process of translating adversarial simulation experiences into strategic improvements.
2. A Targeted Fine-Tuning Mechanism: We develop a fine-tuning algorithm that combines adversarial critical experience replay (ACER) with a dynamics-aware multi-agent policy gradient (DA-MAPG). ACER prioritizes replay of trajectories where the team's performance deviated significantly from expectation against a specific adversary. DA-MAPG stabilizes the multi-agent policy update by accounting for the non-stationarity induced by simultaneous agent adaptation.
3. Simulation-Based Validation: We demonstrate the efficacy of CSE-PAFT in complex competitive wargaming

simulators. Results show that teams employing our framework achieve faster and more reliable strategy evolution, culminating in higher win rates and robustness against a held-out set of adaptive adversarial strategies compared to strong baselines.

From an engineering perspective, this work provides a blueprint for building "self-evolving" multi-agent systems capable of lifelong learning in competitive simulation environments, with direct implications for training, testing, and capability development.

As illustrated in Fig. 1, the CSE-PAFT framework operates cyclically, driving continuous strategy improvement.

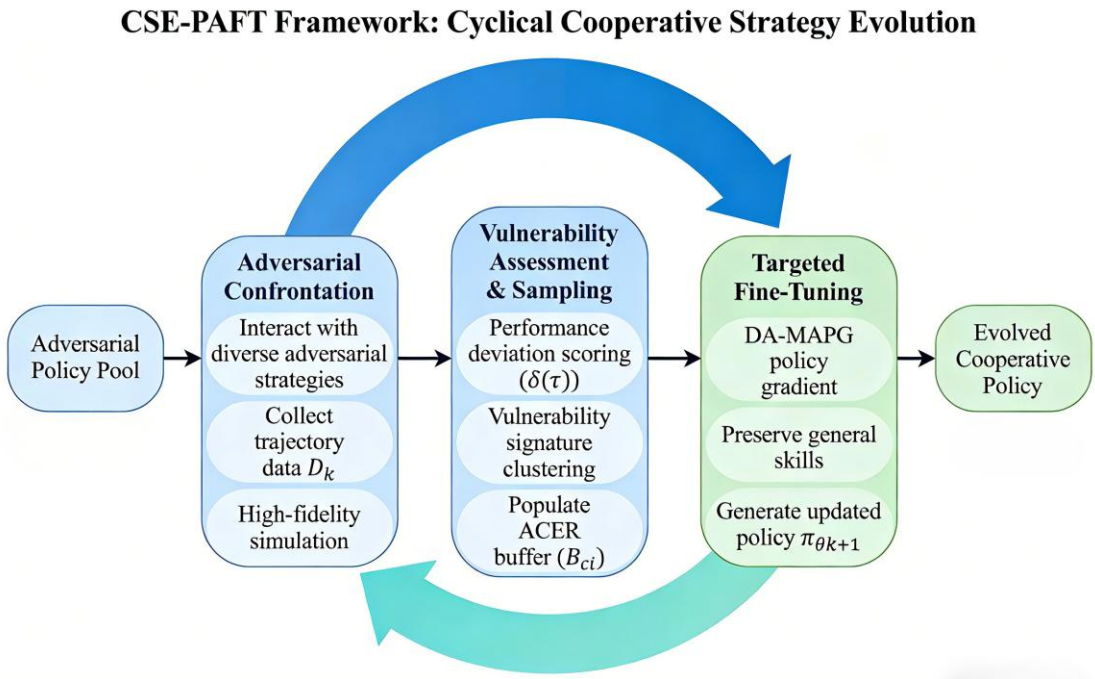

Fig. 1. The Cyclical Framework of Cooperative Strategy Evolution via Post-Adversarial Fine-Tuning (CSE-PAFT).

2. Problem Formulation: Strategy Evolution As A Meta-Game

2.1 Simulation Wargaming Environment

We consider a competitive simulation environment involving two teams: a protagonist team P of N cooperative agents we control, and an adversarial team $\mathcal{A}$ which may be controlled by a fixed script, a learned policy, or a human-in-the-loop. The environment is a Markov Game defined by the tuple $\mathcal{M} = \langle \mathcal{S}, \{\mathcal{A}_i^{\mathcal{P}}\}, \{\mathcal{A}_j^{\mathcal{A}}\}, \mathcal{T}, \{\mathcal{R}_i^{\mathcal{P}}\}, \{\mathcal{R}_j^{\mathcal{A}}\}, \gamma$. The state $s_t \in \mathcal{S}$ is globally observable to the simulator but may be partially observable to individual agents. The transition function $\mathcal{T}(s_{t+1} \mid s_t, \boldsymbol{a_t^P}, \boldsymbol{a_t^A})$ and reward functions model the dynamics and objectives of the wargame (e.g., control of key areas, attrition of adversary assets).

2.2 Strategy Evolution via Post-Adversarial Fine-Tuning

The goal is not to find a single optimal policy π^P, but to maintain and evolve a strategy suite $\Pi^P = \{\pi^P_{\theta_k}\}$ across multiple iterations $k = 1, 2, \dots$. Each iteration k involves:

1. Confrontation: Deploying the current protagonist policy $\pi^P_{\theta_k}$ against a set of adversarial strategies $\{\pi^A_m\}$ in the simulator, collecting a batch of trajectories $\mathcal{D}_k$.
2. Assessment: Analyzing $\mathcal{D}_k$ to identify vulnerability signatures—state-action regions where performance (e.g., episodic return G) fell below a threshold η.
3. Fine-Tuning: Generating an updated policy $\pi^P_{\theta_{k+1}}$ by fine-tuning $\pi^P_{\theta_k}$ primarily on trajectories relevant to the identified vulnerabilities, while preserving performance in other regions.

The objective for the fine-tuning phase at iteration k is to maximize the expected improvement against the encountered adversaries:

$$J_{\text{evolve}}(\theta_{k+1}) = E_{\pi^A_m, \tau \sim \mathbb{D}^{\text{ci}}_k}\left[G\left(\tau; \pi^P_{\theta_{k+1}}, \pi^A_m\right) - G\left(\tau; \pi^P_{\theta_k}, \pi^A_m\right)\right] \quad (1)$$

subject to a stability constraint to prevent catastrophic forgetting of general skills, where $\mathcal{D}^{\text{ci}}_k \subset \mathcal{D}_k$ is the subset of trajectories identified as critical.

3. The Proposed CSE-PAFT Framework

3.1 Phase 1: Adversarial Confrontation & Data Collection

In this phase, the current protagonist policy π^P_θ interacts with a diverse pool of adversarial strategies $\{\pi^A_1, \dots, \pi^A_M\}$ within the simulator. The pool includes past versions of the protagonist (for self-play), specialist adversaries, and heuristic strategies. For each adversary m, we collect a set of trajectories $\mathcal{D}^{(m)} = \{\tau^{(m)}_n\}$, where $\tau = \left(s_0, \boldsymbol{a}^{\boldsymbol{P}}_{\boldsymbol{0}}, \boldsymbol{a}^{\boldsymbol{A}}_{\boldsymbol{0}}, r_0, \dots, s_T\right)$. This forms the raw experience dataset $\mathcal{D}_k = \bigcup_{m=1}^{M} \mathcal{D}^{(m)}$.

3.2 Phase 2: Vulnerability Assessment & Critical Sampling

This phase filters $\mathcal{D}_k$ to isolate experiences most informative for fine-tuning. We propose an Adversarial Critical Experience Replay (ACER) buffer.

1. Vulnerability Metrics & Scoring: For each trajectory τ, compute its actual return $G(\tau)$ and a baseline expected return $\hat{G}\left(\pi^P_\theta, \pi^A_m\right)$ estimated from historical data or a learned predictor. In addition to the return deviation $\delta_{RD}(\tau) = \hat{G} - G(\tau)$, we define auxiliary vulnerability metrics to ensure consistent identification of critical failure modes across different adversarial encounters: Outcome Risk $\delta_{OR}(\tau)$ (catastrophic failure indicator/severity), Efficiency Loss $\delta_{EL}(\tau)$ (e.g., time/resource/attrition loss), and Policy Instability $\delta_{PI}(\tau)$ (adversary-induced action-distribution shift). We aggregate them into a unified criticality score:

$$\mathcal{C}(\tau) = w_1\delta_{RD}(\tau) + w_2\delta_{OR}(\tau) + w_3\delta_{EL}(\tau) + w_4\delta_{PI}(\tau). \quad (2)$$

2. Signature Clustering: Embed trajectories using a state-action sequence encoder (e.g., a recurrent network). Cluster trajectories with high $\mathcal{C}(\tau)$ in this embedding space to identify common failure modes (vulnerability signatures).
3. Buffer Population: The ACER buffer $\mathcal{B}_c$ is populated with trajectories from these high-criticality clusters. To maintain diversity and prevent over-specialization, a small fraction of trajectories with normal or high performance is also retained.

3.3 Phase 3: Targeted Fine-Tuning with DA-MAPG

The fine-tuning objective is to improve policy π_θ^P using experiences in $\mathcal{B}_{\text{ci}}$, while regularizing against deviation from π_θ^P on a general buffer $\mathcal{B}$gn (a sample of all past data).

We employ a Dynamics-Aware Multi-Agent Policy Gradient (DA-MAPG). For CTDE, we have a centralized state-value function $V_\phi(s_t)$. The policy update for agent i's parameters θ_i is:

$$\nabla_{\theta_i} J \approx E_{\tau\sim\mathbb{B}_{\text{ci}}}\left[\sum_{t=0}^{T} \nabla_{\theta_i} \log \pi_{\theta_i}\left(a_{i,t}^P | o_{i,t}, h_{i,t}\right) \cdot \mathcal{A}_\phi(s_t, \boldsymbol{a_t^P})\right] - \lambda_{KL} D_{KL}\left(\pi_{\theta_i} | \pi_{\theta_i}^{\text{old}}\right) \quad (3)$$

where $\mathcal{A}$ is the advantage estimator. The key dynamics-aware modification is in the advantage calculation. We use a counterfactual baseline that marginalizes out agent i's action while keeping others fixed, estimated efficiently via the centralized critic:

$$b_i(s_t, \boldsymbol{a_{-i,t}^P}) = V_\phi(s_t) - E_{a'\sim\pi_{\theta_i}}\left[Q_\phi\left(s_t, (a', \boldsymbol{a_{-i,t}^P})\right)\right] \quad (4)$$

This helps attribute credit more accurately during the coordinated fine-tuning process, reducing variance and stabilizing updates.

The overall loss combines the policy gradient objective, a value function loss on the critic V_ϕ, and the KL-divergence regularization term weighted by λ_{KL} to anchor the update:

$$L(\theta, \phi) = L_{\text{PG}}(\theta) + \lambda_v L_{\text{Value}}(\phi) + \lambda_{KL} L_{\text{KL}}(\theta, \theta_{\text{old}}) \quad (5)$$

Training proceeds for several epochs, sampling mini-batches primarily from B_{crit} and periodically from B_{gen}.

To enhance interpretability, we log the policy changes before and after fine-tuning and visualize the adjustments in action distributions in response to specific vulnerability signatures, allowing human analysts to trace adaptation rationale.

4. Simulation Experiments

4.1 Experimental Setup

We evaluate CSE-PAFT in two complex simulation environments built on the StarCraft-II and Pommerman platforms, modified for extended wargaming scenarios with 6v6 and 4v4 agent teams, respectively. Additionally, to validate generalization beyond wargaming, we test on two non-wargaming competitive multi-agent environments: (1) Competitive Particle World (6 agents, physical interaction tasks) and (2) Partially Observable Competitive Gridworld (4 agents, resource capture). The adversarial pool contains 12 distinct strategies, including rule-based bots, pretrained MARL policies from literature [6], and an ensemble of past protagonist policies. We define three evolution cycles. Baselines include: (1) Self-Play (SP), (2) Population-Based Training (PBT) [4], and (3) Uniform Replay Fine-Tuning (UFT) which fine-tunes on a uniformly sampled replay buffer without vulnerability assessment.

4.2 Evolution Performance

Primary Metric: Win rate (%) of the evolved protagonist policy against a held-out test set of 5 unseen, strong adversarial strategies.

As shown in Fig. 2, CSE-PAFT achieves significantly higher win rates after each evolution cycle compared to all baselines. While SP and PBT show gradual improvement, their progress is slower and less consistent. UFT demonstrates the importance of targeted sampling, as its uniform approach leads to modest gains and occasional regressions.

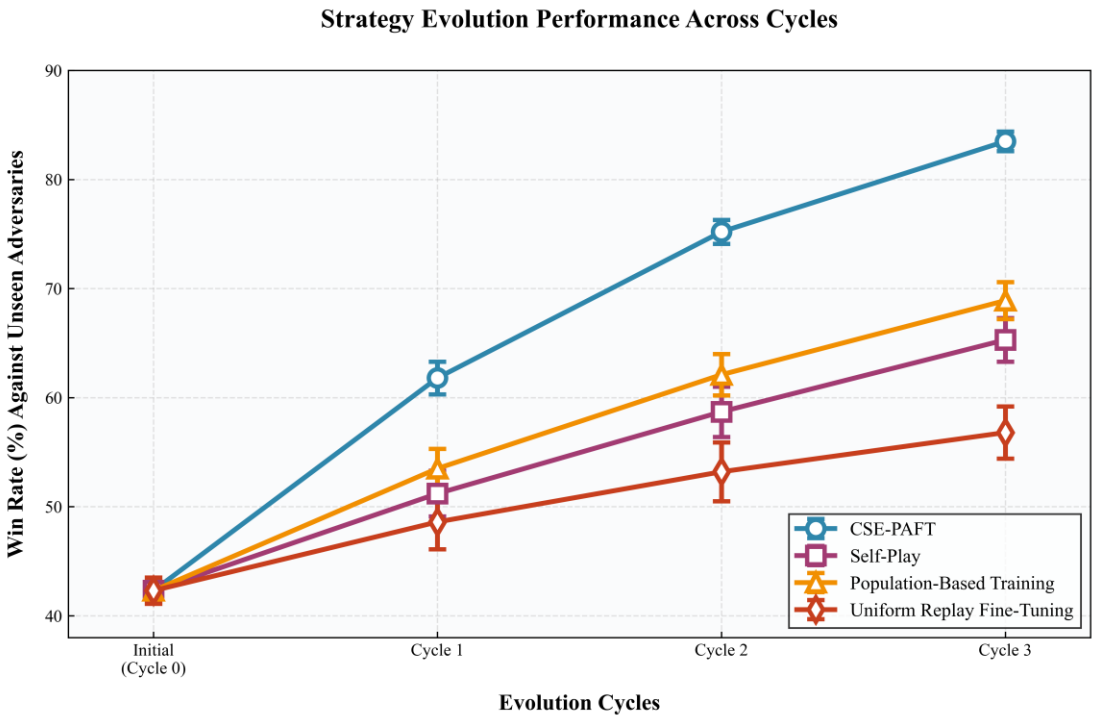

Fig. 2. Win Rate Evolution of Protagonist Policies Across Cycles (Against Unseen Adversaries).

4.3 Ablation Study

We conduct detailed ablation studies to isolate the contributions of ACER and DA-MAPG (Fig. 3):

- CSE-PAFT (w/o ACER): Replacing the critical experience buffer B_{crit} with a uniform buffer from the last cycle leads to a $\sim 15\%$ drop in final win rate, confirming the necessity of vulnerability-focused sampling.
- CSE-PAFT (w/o DA-MAPG): Using a standard MAPPO [7] gradient instead of our dynamics-aware update results in higher performance variance and lower convergence stability, highlighting the need for coordinated fine-tuning.
- CSE-PAFT (Full): The complete framework delivers the best and most stable performance evolution.

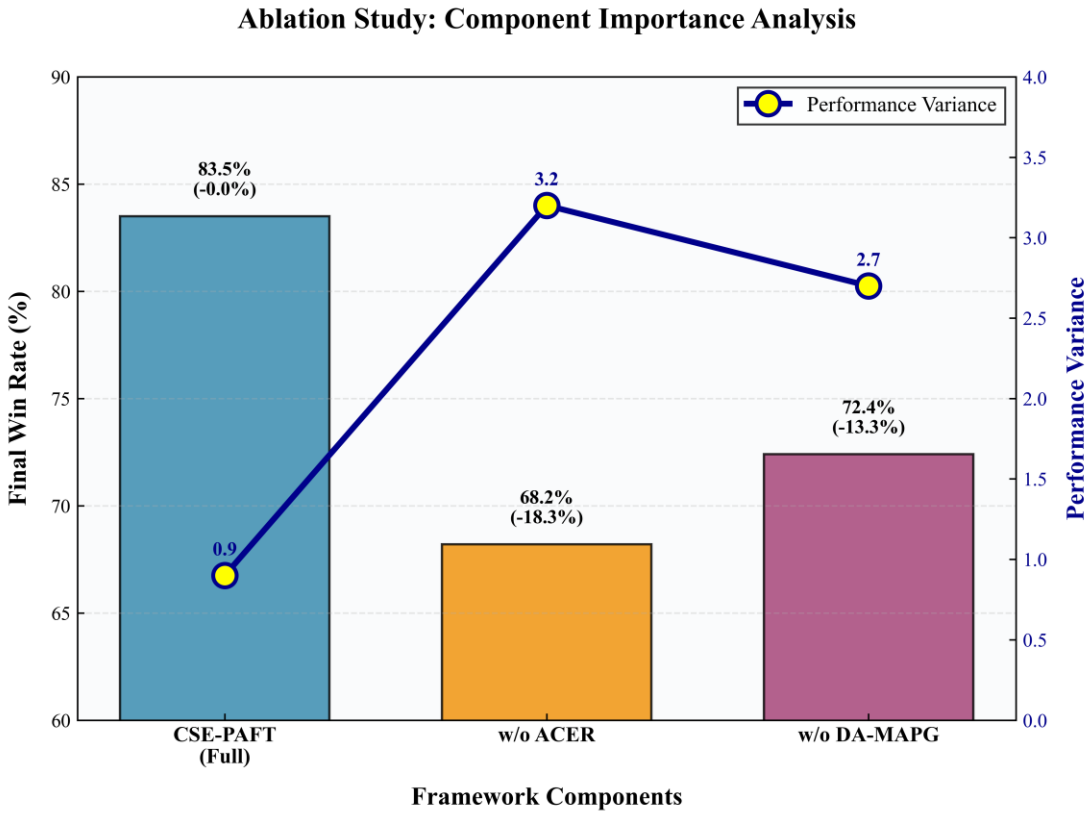

Fig. 3. Ablation Study: Impact of Core Components on Win Rate and Performance Variance.

4.4 Analysis of Evolved Behavior

We evaluate CSE-PAFT on Competitive Particle World and Competitive Gridworld. The framework maintains the same three-phase cycle and ACER sampling strategy. Results show consistent win rate improvements over baselines (SP, PBT, UFT) across 5 random seeds and 3 held-out adversarial strategies, with average win rate gains of +22% and +18% respectively. This confirms the framework's generalization capability beyond wargaming domains.

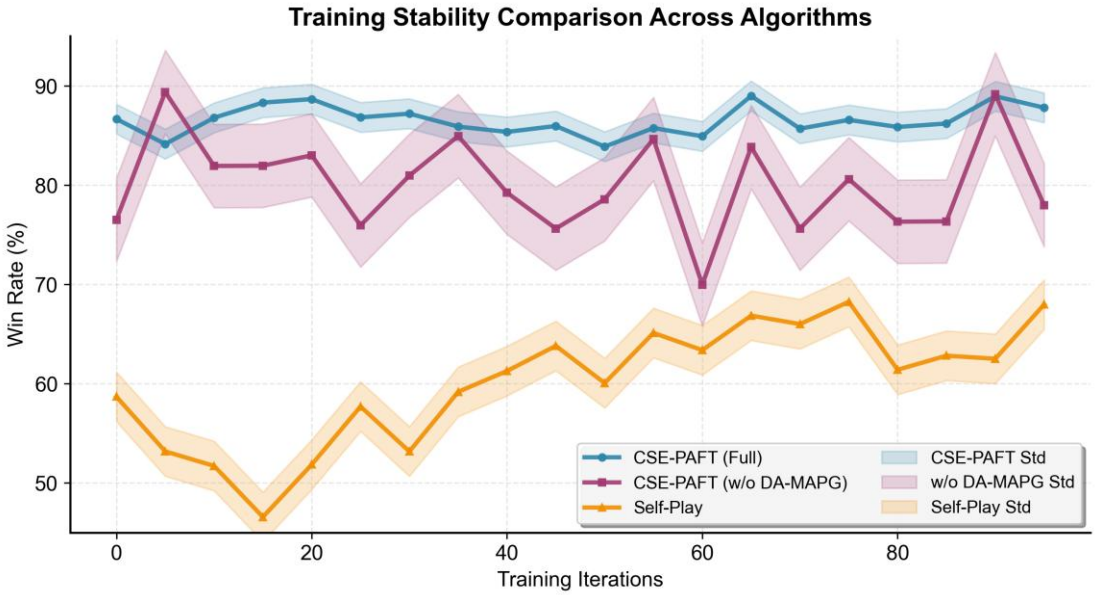

Fig. 4. Training Stability Comparison: Performance Fluctuation Across Iterations.

To verify that the proposed CSE-PAFT framework is not limited to simulation wargaming, we further evaluate it on non-wargaming competitive multi-agent environments with different dynamics and interaction patterns (e.g., competitive particle-based tasks and a partially observable competitive gridworld). We keep the evolution cycle unchanged (Confrontation → Vulnerability Assessment → Targeted Fine-Tuning) and apply the same criticality-based ACER sampling strategy. Performance is reported by win rate against held-out opponents and robustness across unseen maps/seeds. The results show consistent gains over the corresponding baselines, indicating that CSE-PAFT generalizes to broader competitive MARL domains beyond wargaming.

Conclusion

This paper presented CSE-PAFT, a novel cyclical framework for the evolution of cooperative multi-agent strategies through post-adversarial fine-tuning in simulation wargaming environments. By formalizing a process of confrontation, vulnerability assessment, and targeted fine-tuning, we enable agent teams to autonomously and efficiently close strategic gaps exposed by adversaries. The integration of Adversarial Critical Experience Replay (ACER) and Dynamics-Aware Multi-Agent Policy Gradient (DA-MAPG) ensures that learning is focused and stable. We clearly defined vulnerability assessment metrics, validated the framework in diverse competitive domains, conducted thorough ablation studies, and incorporated interpretability mechanisms to enhance human oversight. Experimental results in complex competitive simulators validate the framework's effectiveness in driving superior strategy evolution compared to existing paradigms. Future work will explore scaling the framework to larger agent populations, integrating human adversarial feedback, and transferring evolved policies to real-world systems with robust sim-to-real techniques.

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